PINGCHENG JIAN

PhD student, Electrical and Computer Engineering, Duke University
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EDUCATION

Duke University *Sep. 2021 - present*
Ph.D. in Electrical and Computer Engineering Advisors: Michael Zavlanos, Boyuan Chen

Tsinghua University *Aug. 2016 - Jul. 2021*
B.E. in Automation Advisors: Huaping Liu, Mingguo Zhao
Transfer major from Mechanical Engineering (2016-2018)
to Automation (2018-2021)

RESEARCH INTERESTS

My area of research is robot learning, and my previous publications cover the 3 topics: 1) applying foundation models for the preference feedback in reinforcement learning, 2) modularity for robot learning, and 3) adversarial learning for robust control.

PUBLICATIONS

(∗ equal contributions, ∗∗ alphabetical order. Up-to-date list can be found in my Google Scholar page.)

Journal Papers

1. **[TMLR'24]** Pingcheng Jian, Easop Lee, Zachary I. Bell, Michael M. Zavlanos, Boyuan Chen, "Perception Stitching: Zero-Shot Perception Encoder Transfer for Visuomotor Robot Policies", *Transactions on Machine Learning Research (TMLR)*, 2024. [openreview] [website] [video]] [arxiv]

Conference Papers

2. **[RSS'25 in submission]** Pingcheng Jian, Xiao Wei, Yanbaihui Liu, Samuel A. Moore, Michael M. Zavlanos, Boyuan Chen, "LAPP: Large Language Model Feedback for Preference-Driven Reinforcement Learning", *In submission to Robotics: Science and Systems (RSS)*, 2025. [pdf]
3. **[CoRL'23]** Pingcheng Jian, Easop Lee, Zachary Bell, Michael M. Zavlanos, Boyuan Chen, "Policy Stitching: Learning Transferable Robot Policies", *Conference on Robot Learning (CoRL)*, 2023. [website] [CoRL talk] [video] [arXiv]
4. **[ICRA'21]** Pingcheng Jian∗, Chao Yang∗, Di Guo, Huaping Liu, Fuchun Sun, "Adversairial Skil Learning for Robust Manipulation", *International Conference on Robotics and Automation (ICRA)*, 2021. [pdf] [video]
5. **[L4DC'23]** Reza Khodayi-mehr, Pingcheng Jian, Michael M. Zavlanos, "Physics-Guided Active Learning of Environmental Flow Fields", *Learning for Dynamics and Control (L4DC)*, 2023. [pdf]

Workshop Papers

6. **[ICML-W'21]** Minghao Zhang∗, Pingcheng Jian∗, Yi Wu, Huazhe Xu, Xiaolong Wang, "DAIR: Disentangled Attention Intrinsic Regularization for Safe and Efficient Bimanual Manipulation", *International Conference on Machine Learning (ICML) Workshop: Reinforcement Learning for Real Life*, 2021. [website]

PROJECTS

Quadruped Spider Robot, course project of *CS590 - Robot Studio* at Duke University
Project Video: [video]
I built all the hardware and software of this spider robot from scratch. I drew the parts of this robot with SOLIDWORKS and then printed them with 3D printer. The controller of this robot is a Raspberry Pi, and the control algorithm is written in Python. Then I generated the .urdf files of this robot and built the simulator of this robot with PyBullet.

ACADEMIC SERVICE

Reviewer:
Conference on Robot Learning (CoRL), 2024
Conference on Robot Learning (CoRL), 2025

TEACHING EXPERIENCES

Teaching Assistant at Duke University
ECE 382L/ME 344L - Control of Dynamic Systems [lecture] [recitation]
ECE 689/COMPSCI 676 - Advanced Topics in Deep Learning

ADVISING AND MENTORSHIP

Easop Lee, Duke Undergraduate Student in ECE (2022). Next: Ph.D. at Duke ECE
Xiao Wei, Duke CS Master (2024 - present)
Yongjae Lee, Duke ME Master (2024 - present)

SKILLS

Programming Languages
Python, Matlab, C, C++, C#, HTML, Assembly Language
Machine Learning
Pytorch, Reinforcement Learning, Supervised Learning
Robotic Software
PyBullet, MuJoCo, Isaac Gym
Robotic Hardware
SOLIDWORKS, 3D printing